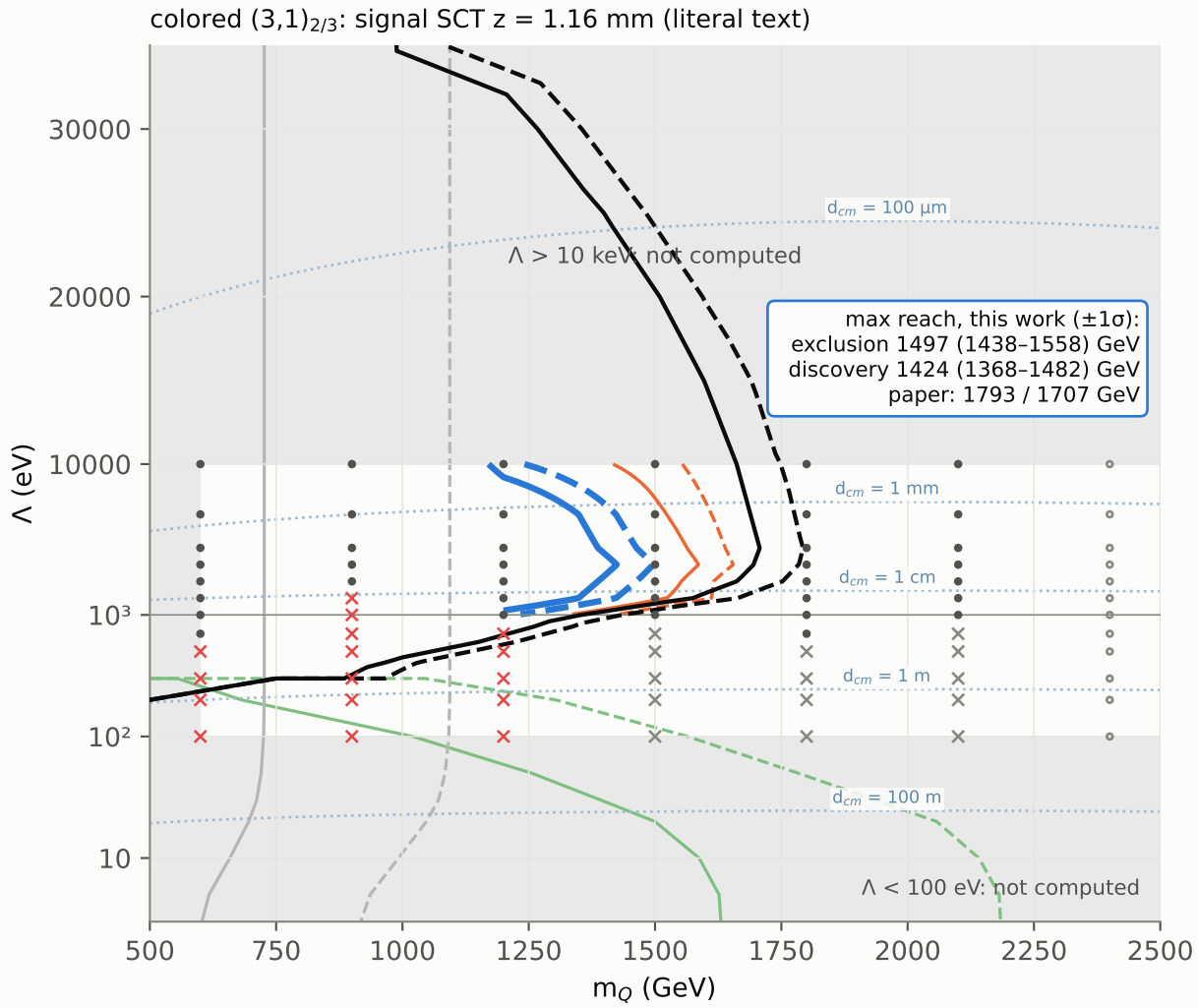
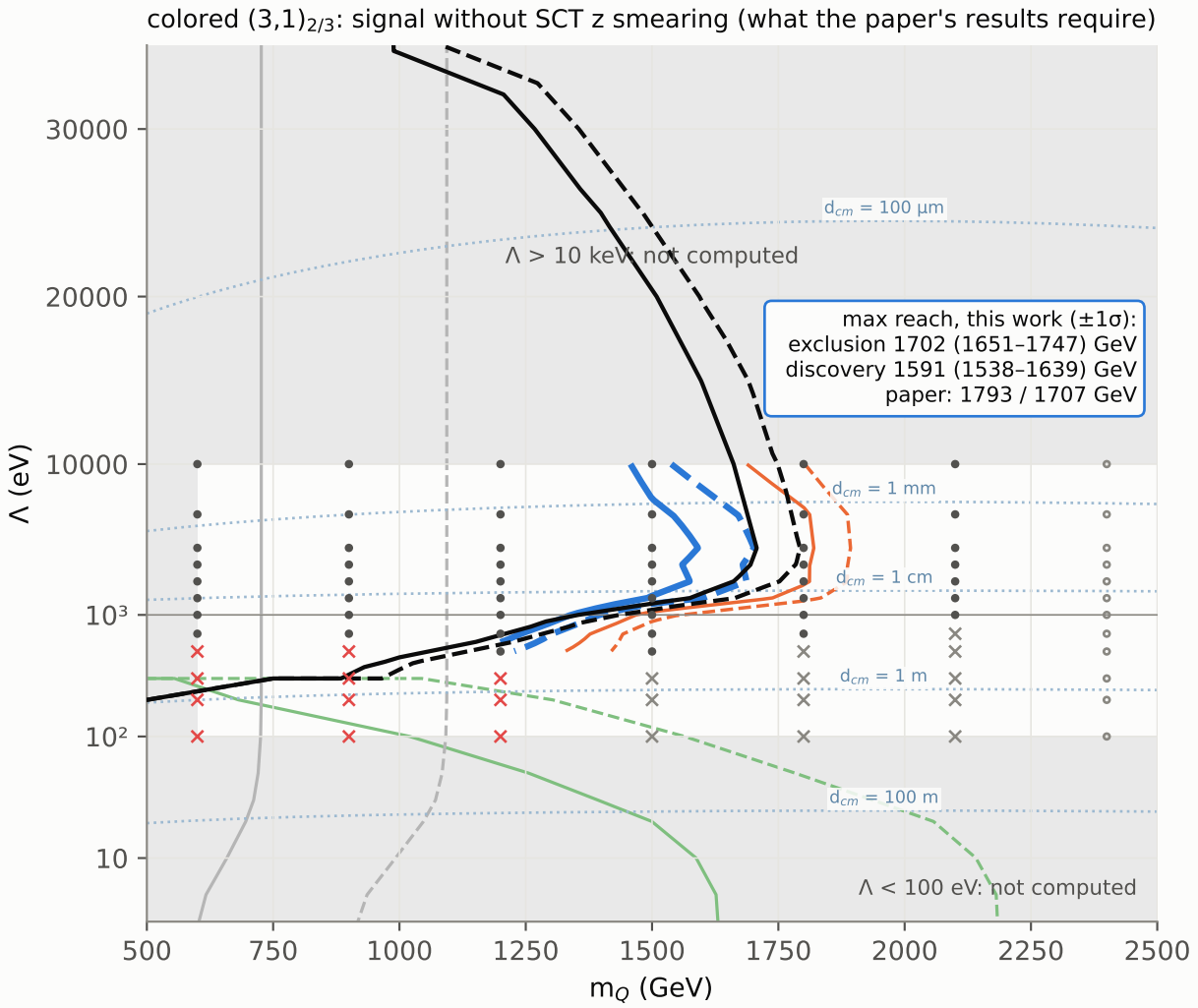
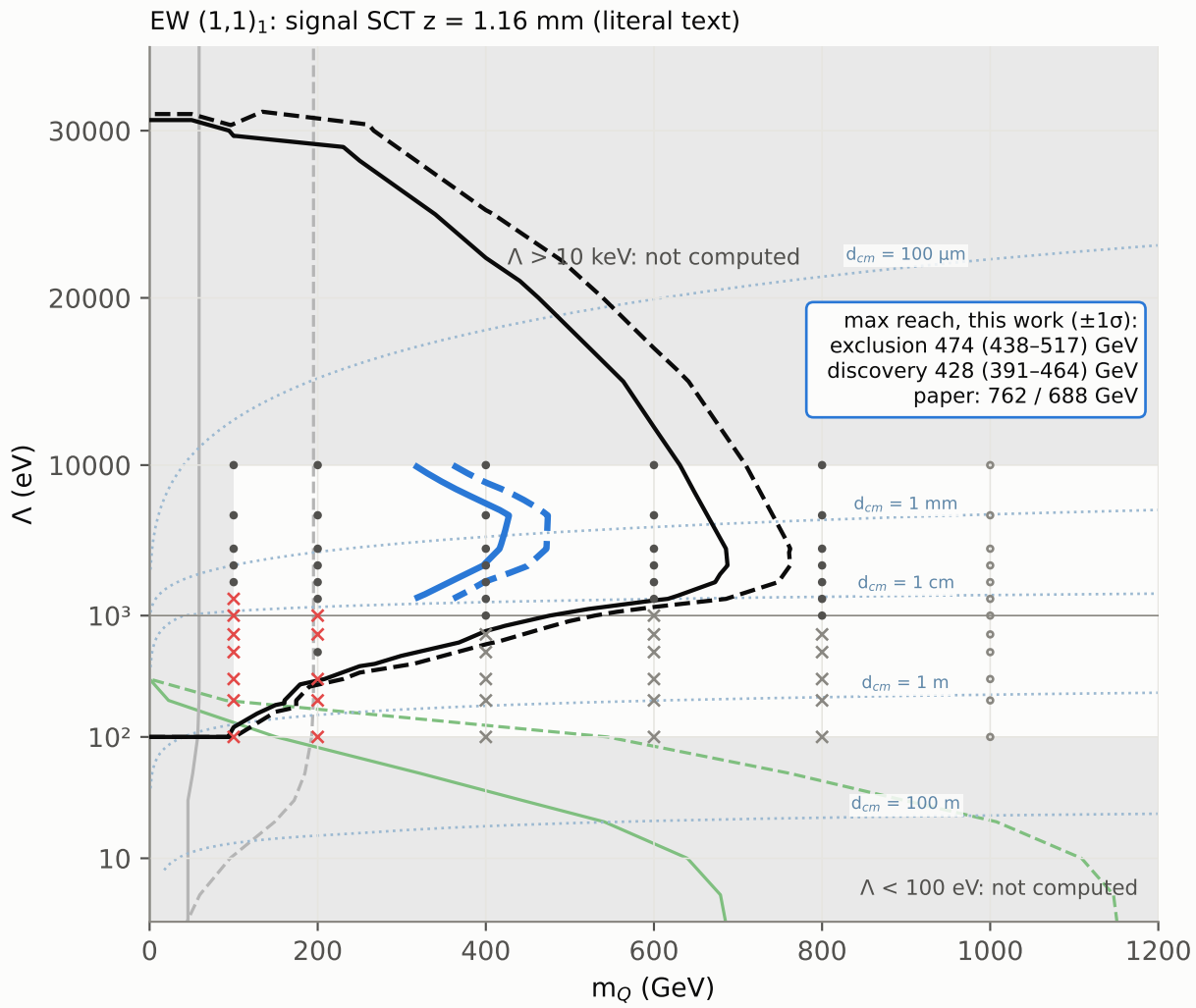
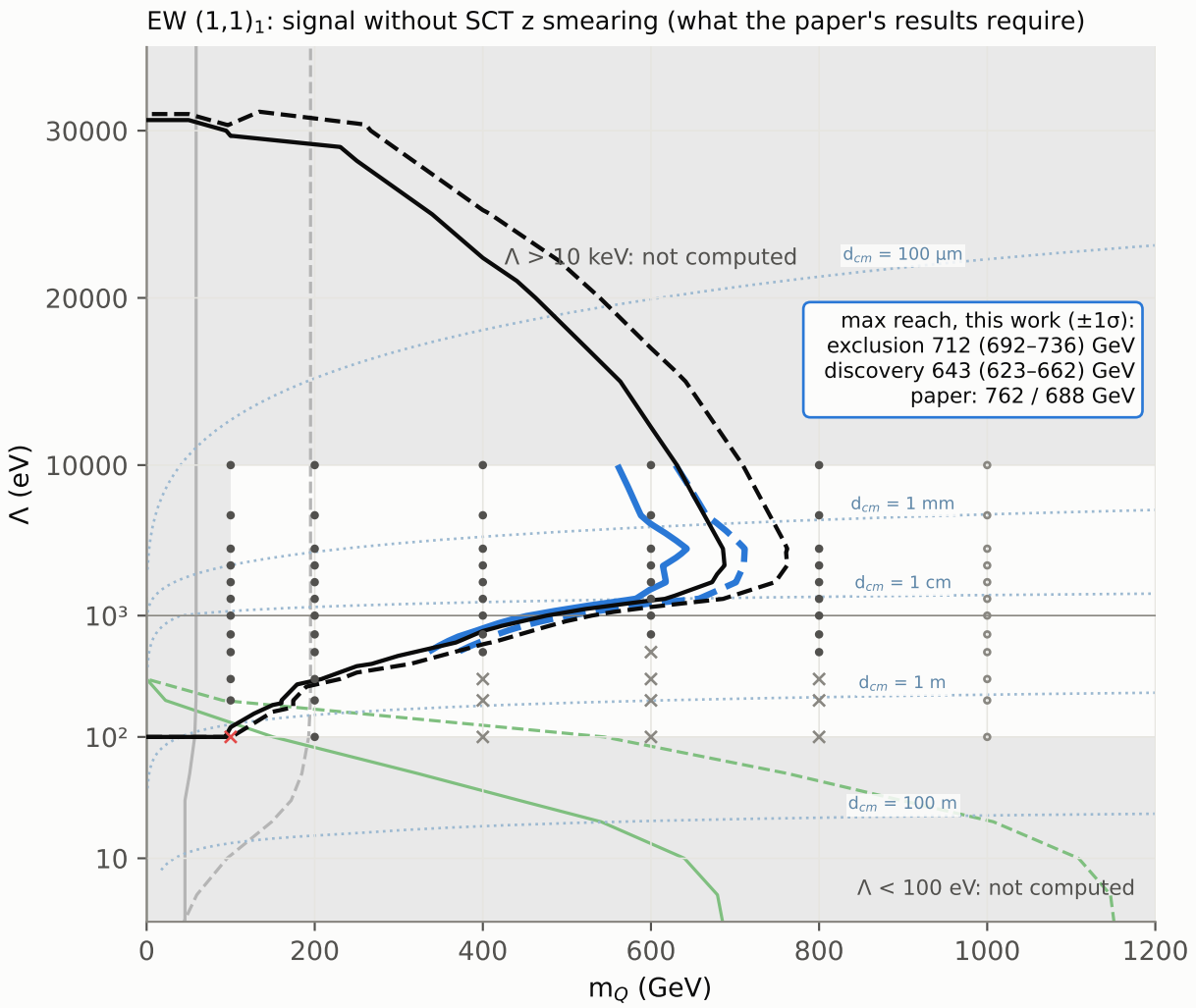
Figure 6 of arXiv:1708.02243: contours of 3 (exclusion) and 5 (discovery) signal events at 300 fb^{-1}, no background, as in the paper
left: signal SCT z = 1.16 mm (literal text); right: signal without SCT z smearing (what the paper's results require)









paper: exclusion (3 events)
paper: discovery (5 events)
this work: exclusion (3 events)
this work: discovery (5 events)
this work, colored, no charge factor: exclusion
this work, colored, no charge factor: discovery
HSCP 12.9 / 300 fb^{-1} (from the paper)
monojet 12.9 / 300 fb^{-1} (from the paper)
d_{cm} (from the paper)
grid point with ≥ 1 event in the signal region
0 events passed: N < 3 at 95% CL
0 events passed: not resolved (95% UL on N ≥ 3)
not propagated: N < 3 even with $\varepsilon_{fid} = \varepsilon_{reco} = 1$

Normalisation: σ = MadGraph5 3.7.3 LO (0 + 1 jet, MLM-matched with Pythia8) for one quirk species, no K-factor; the paper gives no σ. The "N_Q = 2" of the paper's panel titles is NOT applied (no factor 2 on σ). A factor of 1.5–1.7 (colored) and 1.3–1.4 (EW) on σ brings our maxima close to the paper's (sensitivity, not a fit; F6_kfactor).
EW trigger: ε_trig = σ(pp → QQ̄ + j, p_T(j) > 200 GeV) / σ(pp → QQ̄) at LO (0.096 at 800 GeV, = paper 0.10), NOT the matched + shower count (0.178 at 800 GeV), which would move the EW maximum outward by ~60-80 GeV. EW ε_fid and ε_reco: from the matched + shower sample of each mass; the LO sample at 800 GeV (D41, 3000 events) gives ε_fid 0.30 vs 0.31 here and k/n within ±13%, which moves the EW maximum by +5 GeV. Colored: ε_trig = matched count; the both-charged factor (~0.33) is included (thin orange: without it).
Red ×: at low Λ for colored 600–1200 GeV and EW 100–200 GeV our samples (40–70 events per mass) are too small; the paper's lower-left edge passes there and is not tested here (larger samples needed). Λ > 10 keV: not computed (the maxima are at 3–5 keV, so the reach is unaffected; only the upper part of the shape is missing).
Annihilation is not simulated (as in the paper). Worst case (α = 0.05, annihilation chance at every second meeting) it lowers the signal near the maxima by ~17–24% (EW 4–5 keV) and ~17% (colored 5 keV), moving the maxima inward by ~30 GeV (colored) and ~25–40 GeV (EW).